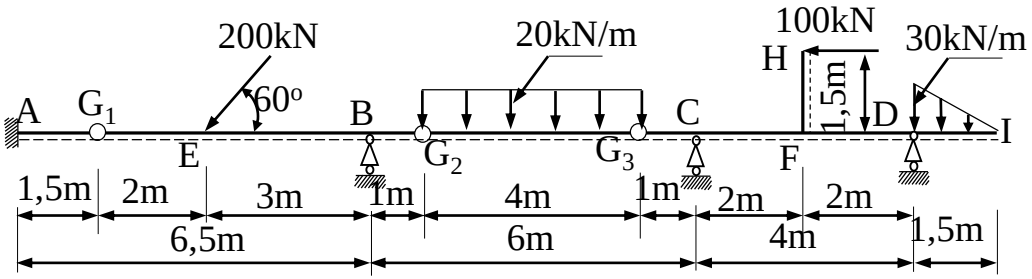
GERBER BEAM EXAMPLES

Example 1:



Draw internal force diagrams of the given system.

Solution:

Components of the inclined load:

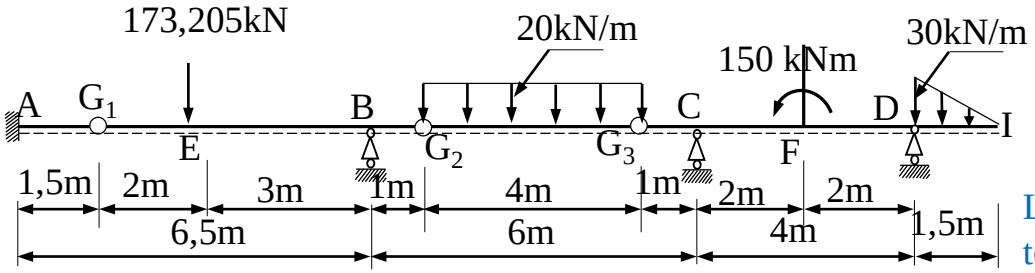
$P_{Ex} = 200 \cdot \cos 60 = 100 \ \ kN$ $\qquad$ $P_{Ey} = 200 \cdot \sin 60 = 173{,}205 \ \ kN$

FH member can be removed from the system and internal forces of the F section of FH member can be applied to the system as external loads:

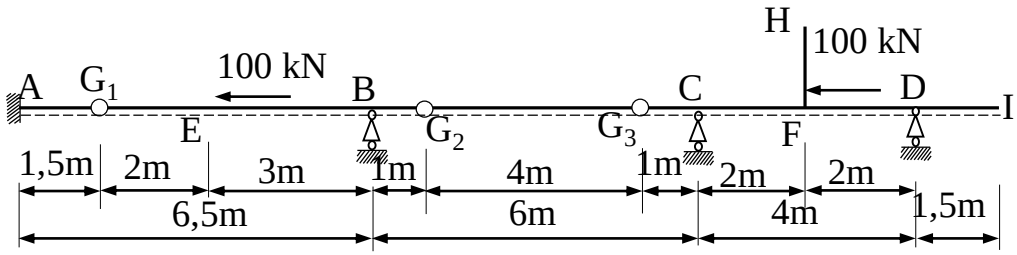
$V_{FH} = -100 \ \ kN$

$M_{FH} = 100 \times 1{,}5 = 150 \ \ kNm$

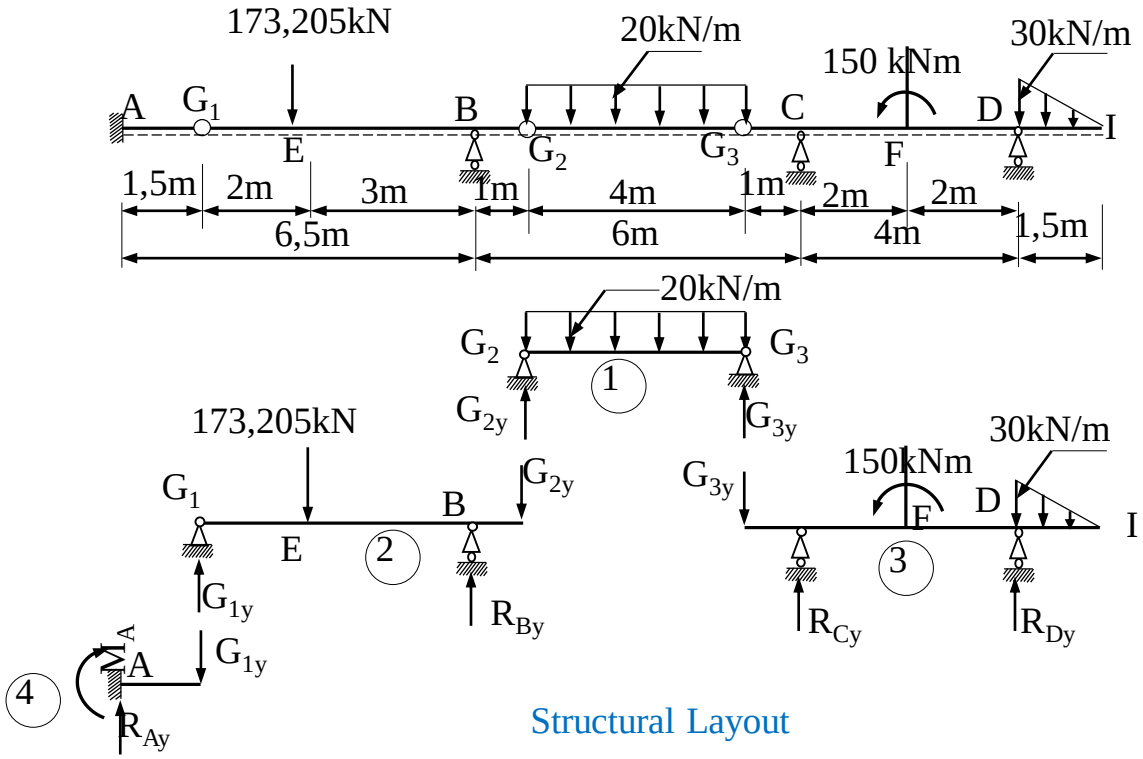
$N_{FH} = 0$



Loads perpendicular to the member axis

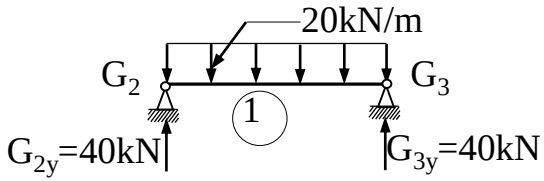


Loads parallel to the member axis



Structural Layout

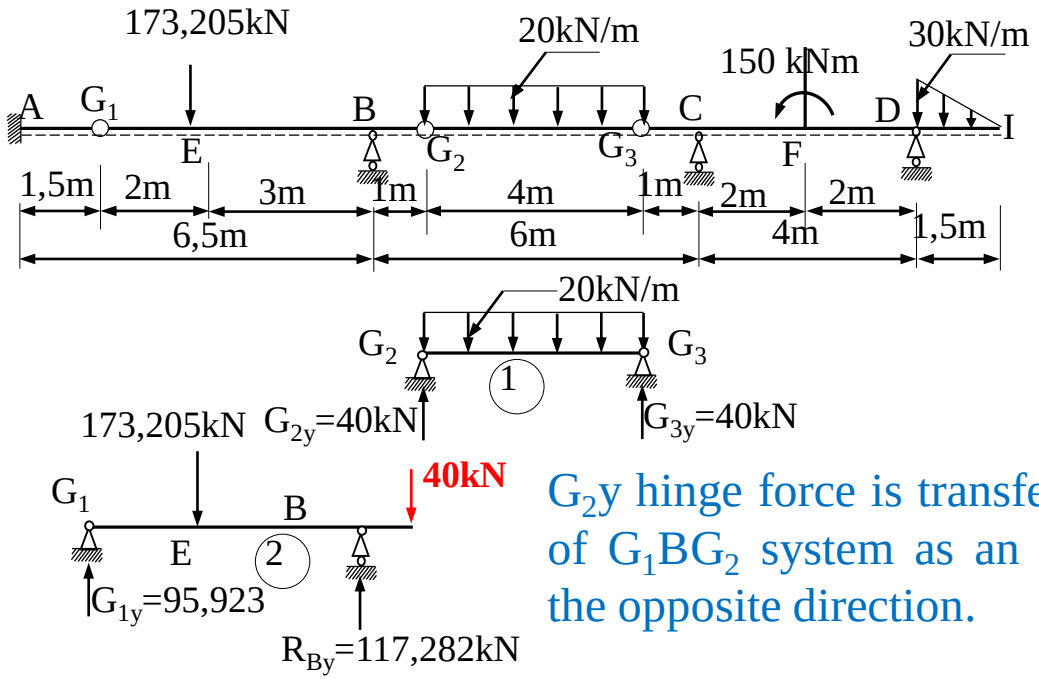
Firstly, G_2G_3 system is solved:



$$G_{2y} = G_{3y} = 20 \times 4{,}0 / 2 = 40 \ \ kN \qquad V_{G2G3} = -V_{G3G2} = 40 \ \ kN$$

Maximum moment at mid-span of Part 1

$$M_{enb.} = 20 \times 4{,}0^2 / 8 = 40 \ \ kNm$$



G_2y hinge force is transferred to point G_2 of G_1BG_2 system as an external load in the opposite direction.

$$\sum M_B = 0 \qquad G_{1y} = (173{,}205 \times 3{,}0 - 40 \times 1{,}0) / 5{,}0 = 95{,}923 \ \text{kN}$$

$$\sum M_{G1} = 0 \qquad R_{By} = (173{,}205 \times 2{,}0 + 40 \times 6{,}0) / 5{,}0 = 117{,}282 \ \text{kN}$$

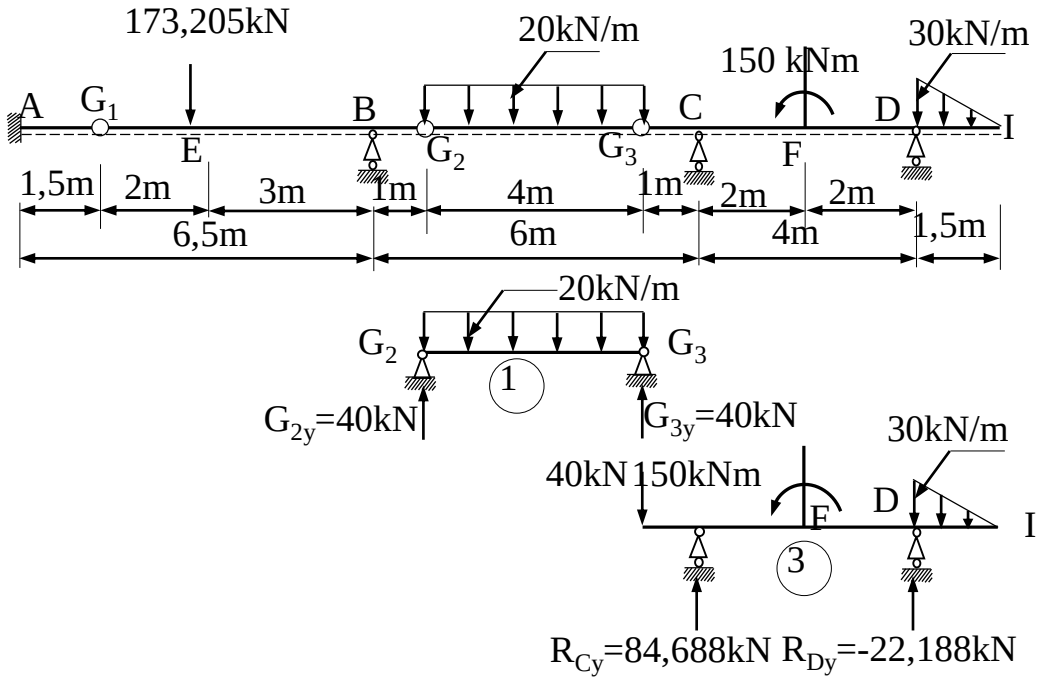
Check: $\sum P_y = 95{,}923 - 173{,}205 + 117{,}282 - 40 = 0$

$$V_{G1E} = V_{EG1} = 95{,}923 \ \text{kN}$$

$$M_{EG1} = M_{EB} = 95{,}923 \times 2{,}0 = 191{,}85 \ \text{kNm}$$

$$V_{EB} = V_{BE} = 95{,}923 - 173{,}205 = -77{,}282 \ \text{kN}$$

$$M_{BE} = M_{BG2} = -40 \times 1{,}0 = -40 \ \text{kNm} \qquad V_{BG2} = V_{G2B} = 40 \ \text{kN}$$



$$\sum M_D = 0 \qquad R_{Cy} = (40 \times 5{,}0 + 150 - 0{,}5 \times 30 \times 1{,}5 \times 0{,}5) / 4 = 84{,}688 \ \text{kN}$$

$$\sum M_C = 0 \qquad R_{Dy} = (-40 \times 1{,}0 - 150 + 0{,}5 \times 30 \times 1{,}5 \times 4{,}5) / 4 = -22{,}188 \ \text{kN}$$

Check: $\sum P_y = -40 + 84{,}688 - 22{,}188 - 22{,}50 = 0$

$$V_{G3C} = V_{CG3} = -40 \ \text{kN} \qquad M_{G3C} = 0$$

$$M_{CF} = M_{CG3} = -40 \times 1{,}0 = -40 \ \text{kNm}$$

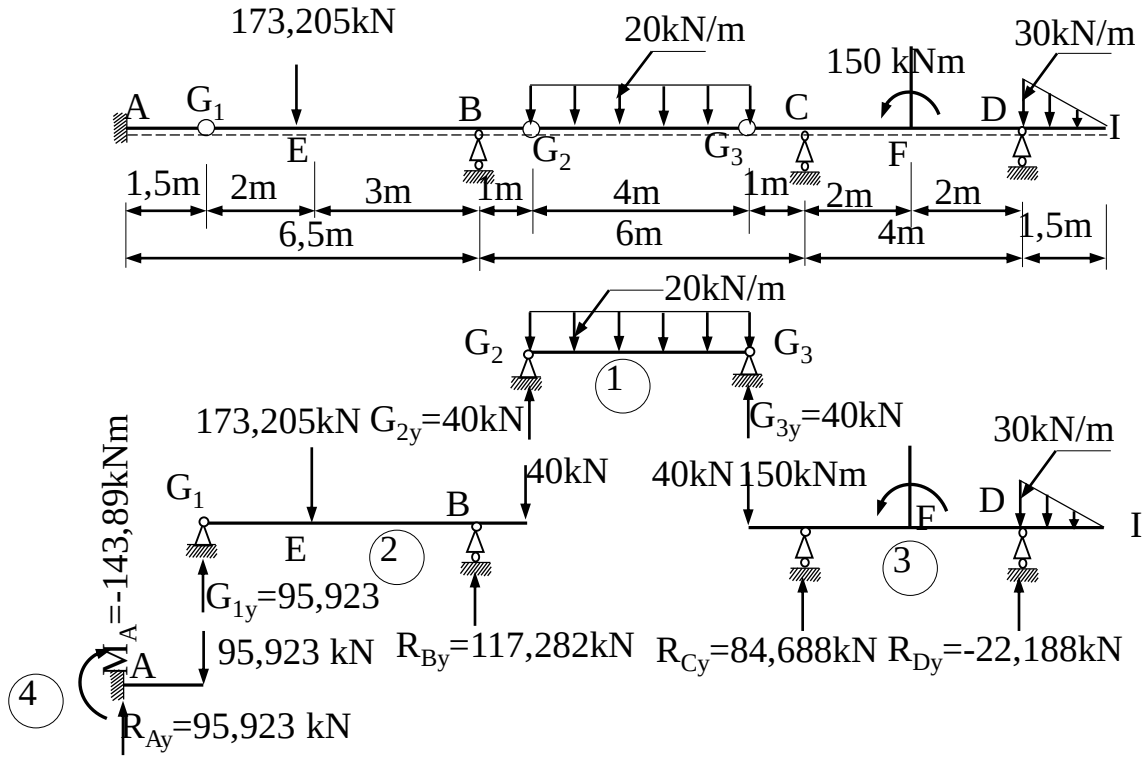
$$V_{CF} = V_{FC} = V_{FD} = V_{DF} = -40 + 84{,}688 = 44{,}688 \ \text{kN}$$

$$M_{FC} = -40 \times 3{,}0 + 84{,}688 \times 2{,}0 = 49{,}38 \ \text{kNm}$$

$$M_{FD} = (-22{,}188) \times 2{,}0 - 22{,}5 \times 2{,}5 = -100{,}62 \ \text{kNm}$$

$$M_{DF} = M_{DI} = -22{,}5 \times 0{,}5 = -11{,}25 \ \text{kNm}$$

$$V_{DI} = 22{,}50 \ \text{kN} \qquad V_{ID} = 0 \qquad M_{ID} = 0$$



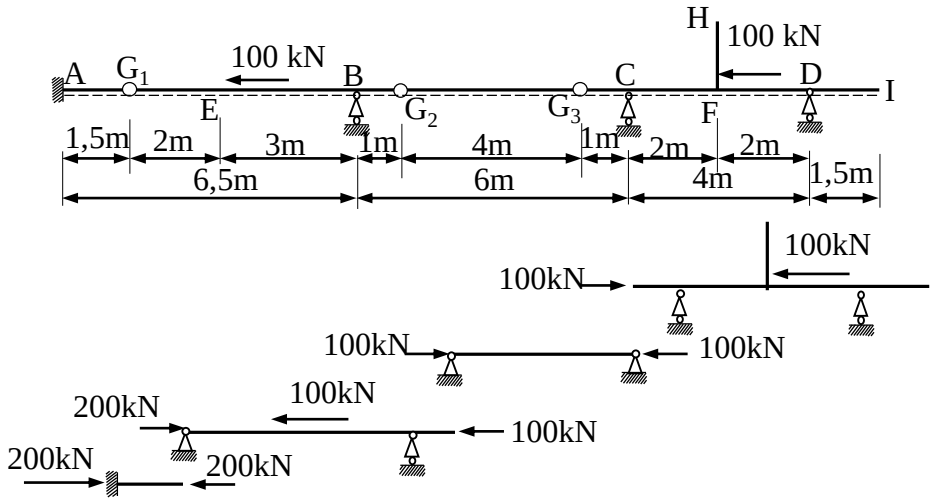
$$\sum P_y = 0 \qquad R_{Ay} = 95{,}923 \ \text{kN}$$

$$\sum M_A = 0 \qquad M_A = -95{,}923 \times 1{,}5 = -143{,}89 \ \text{kNm}$$

$$V_{AG1} = V_{G1A} = 95{,}923 \ \text{kN}$$

$$M_{AG1} = -95{,}923 \times 1{,}5 = -143{,}89 \ \text{kNm}$$

Solution of the system due to the external loads parallel to member axis



$$\sum P_x = 0 \qquad G_{3x} = 100 \ \text{kN}$$

$$\sum P_x = 0 \qquad G_{2x} = 100 \ \text{kN}$$

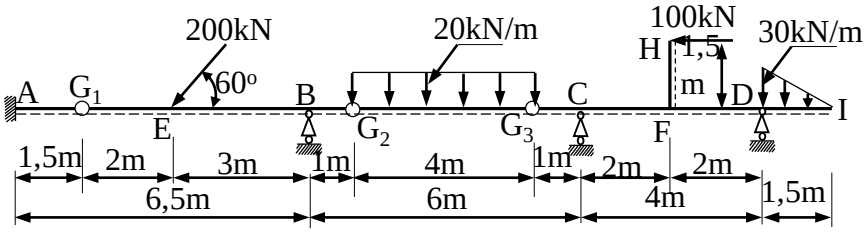
$$\sum P_x = 0 \qquad G_{1x} = 100 + 100 = 200 \ \text{kN}$$

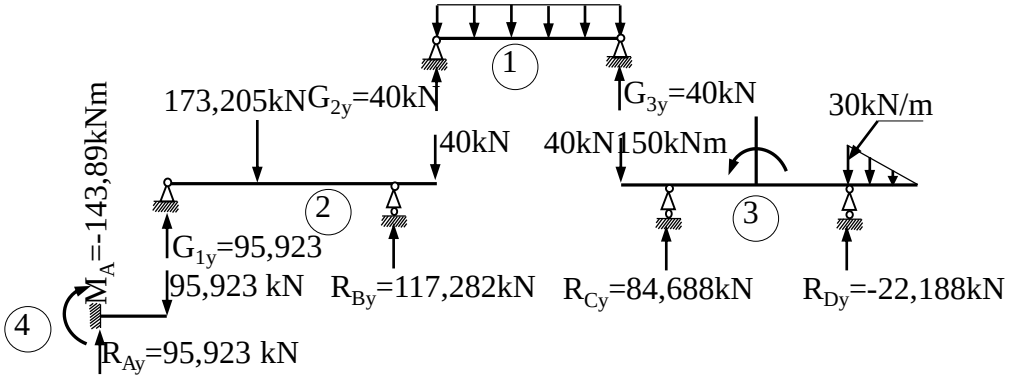
$$\sum P_x = 0 \qquad R_{Ax} = 200 \ \text{kN}$$

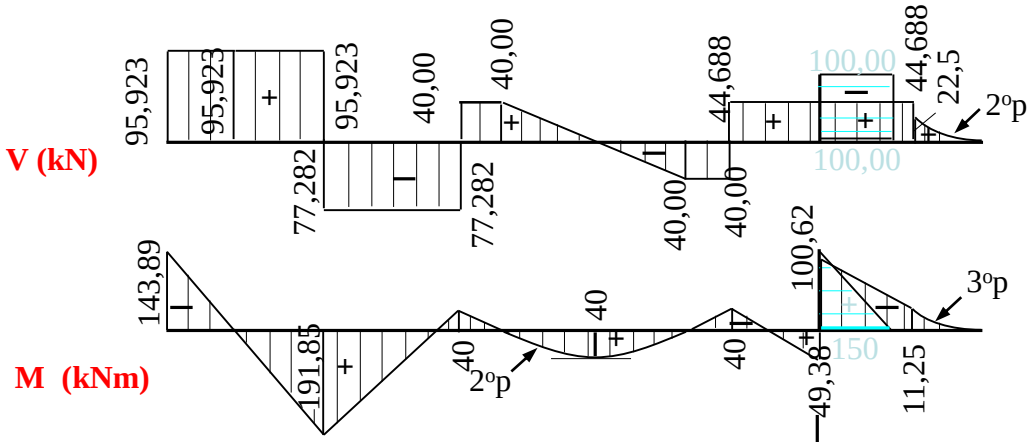
$$N_{FD} = N_{DF} = N_{DI} = N_{ID} = 0 \qquad N_{FC} = N_{CF} = N_{CG3} = N_{G3C} = -100 \ \text{kN}$$

$$N_{G2G3} = N_{G3G2} = -100 \ \text{kN} \qquad N_{G2B} = N_{BG2} = N_{BE} = N_{EB} = -100 \ \text{kN}$$

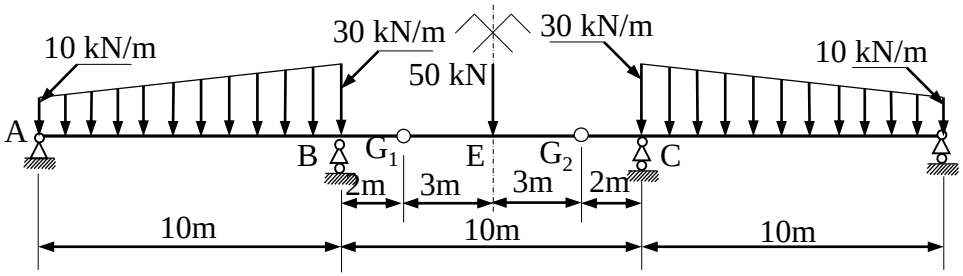
$$N_{EG1} = N_{G1E} = -100 - 100 = -200 \ \text{kN} \qquad N_{AG1} = N_{G1A} = -200 \ \text{kN}$$


200kN
60°
20kN/m
100kN
1,5 m
30kN/m
A
G1
B
C
H
D
I
E
G2
G3
F
1,5m
2m
3m
1m
4m
1m
2m
2m
1,5m
6,5m
6m
4m


1
173,205kN
G2y=40kN
G3y=40kN
30kN/m
40kN
40kN
150kNm
2
3
MA=-143,89kNm
G1y=95,923
95,923 kN
RBy=117,282kN
RCy=84,688kN
RDy=-22,188kN
4
RAy=95,923 kN

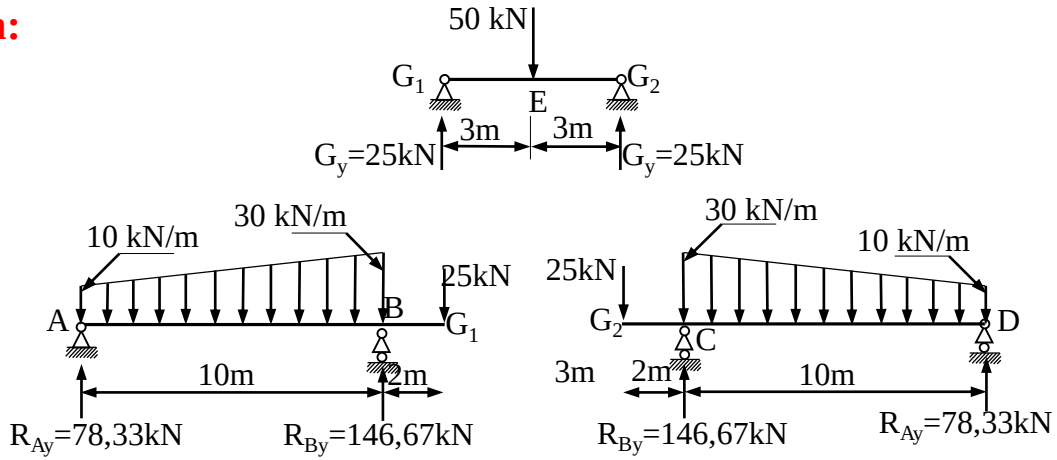

V (kN)
95,923
95,923
95,923
40,00
40,00
44,688
100,00
44,688
22,5
2°p
77,282
77,282
40,00
40,00
100,00
M (kNm)
143,89
191,85
40
2°p
40
40
100,62
3°p
150
49,38
11,25
N (kN)
200
200
100
100
100

Example 2:



Draw internal force diagrams of the given system.

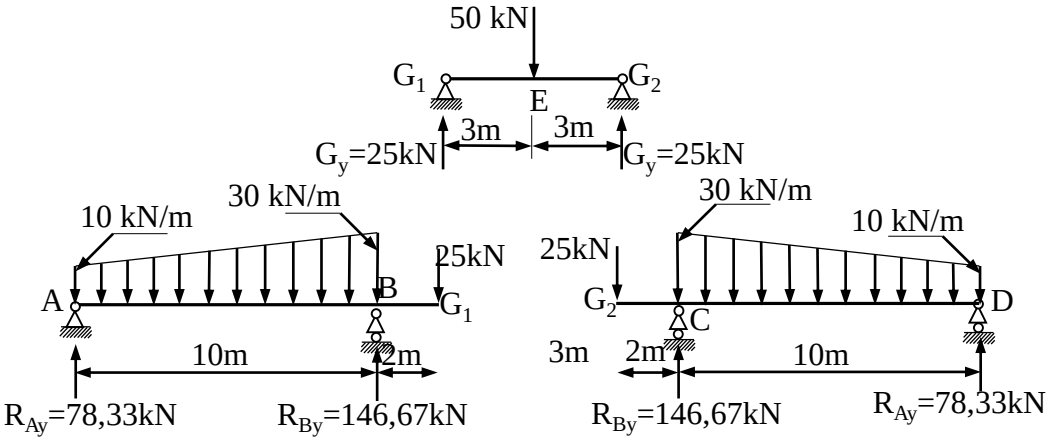
Solution:



G_1G_2 system

$$V_{G1E} = V_{EG1} = 25 \ \text{kN} \qquad G_{1y} = G_{2y} = 50/2 = 25 \ \text{kN}$$

$$M_{EG1} = M_{EG2} = PL/4 = 50 \times 6{,}0/4 = 75 \ \text{kNm}$$



ABG_1 system

Support Reactions:

$$\sum M_B = 0$$

$$R_{Ay} = [10 \times 10{,}0 \times 5 + 0{,}5(30-10) \times 10 \times (10/3) - 25 \times 2{,}0]/10 = 78{,}333 \ \text{kN}$$

$$\sum M_A = 0$$

$$R_{By} = [10 \times 10{,}0 \times 5 + 0{,}5(30-10) \times 10 \times (2 \times 10/3) + 25 \times 12{,}0]/10$$

$$R_{By} = 146{,}667 \ \text{kN}$$

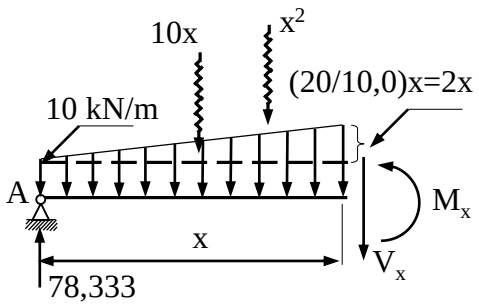
Internal Forces:

$$V_{AB} = 78{,}333 \ \text{kN} \qquad M_{AB} = 0$$

$$V_{BA} = -146{,}667 + 25 = -121{,}667 \ \text{kN}$$

$$M_{BA} = M_{BG1} = -25 \times 2{,}0 = -50 \ \text{kNm}$$

$$V_{BG1} = V_{G1B} = 25 \ \text{kN} \qquad M_{G1B} = 0$$



$$\sum P_y = 0 \longrightarrow \quad V_x = 78{,}333 - 10x - x^2 = 0$$

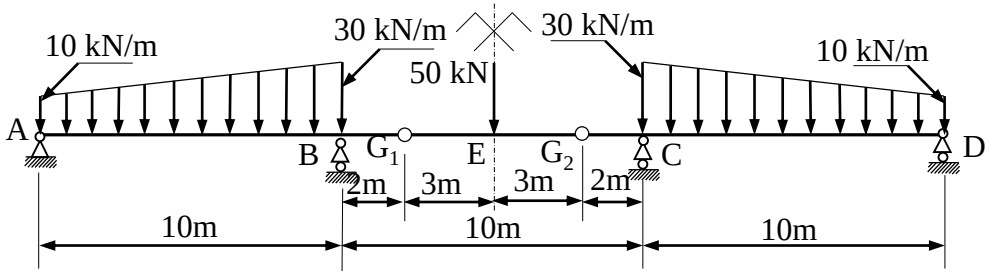
$$\sum M_x = 0 \searrow$$

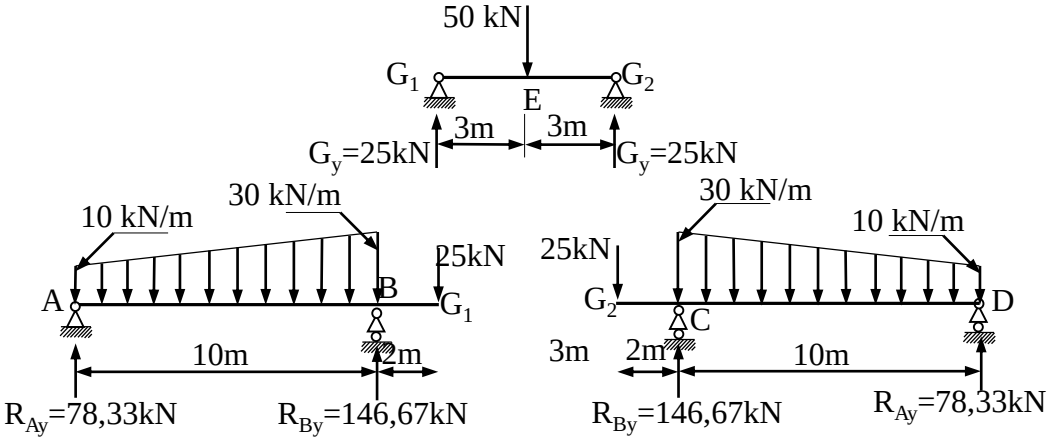
$$M_x = 78{,}333x - 10x \cdot x/2 - x^2 \cdot x/3 = 78{,}333x - 5x^2 - x^3/3$$

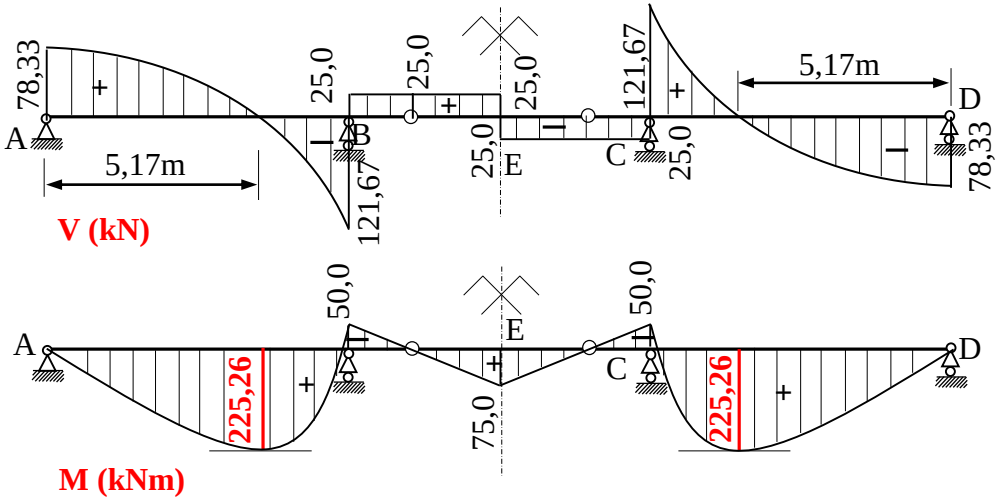
$$M_x = 78{,}333x - 5x^2 - x^3/3$$

$$x = x_0 = 5{,}17 \ \text{m} \searrow$$

$$M_{enb.} = 78{,}333 \times 5{,}17 - 5 \times 5{,}17^2 - 5{,}17^3/3 = 225{,}26 \ \text{kNm}$$


10 kN/m
30 kN/m
30 kN/m
50 kN
10 kN/m
A
B
G1
E
G2
C
D
2m
3m
3m
2m
10m
10m
10m


50 kN
G1
G2
E
3m
3m
Gy=25kN
Gy=25kN
10 kN/m
30 kN/m
25kN
A
B
G1
10m
2m
RAy=78,33kN
RBy=146,67kN
30 kN/m
10 kN/m
25kN
G2
C
D
3m
2m
10m
RBy=146,67kN
RAy=78,33kN


78,33
25,0
25,0
25,0
121,67
5,17m
A
B
E
C
D
5,17m
121,67
25,0
25,0
78,33
V (kN)
50,0
50,0
A
E
C
D
225,26
225,26
75,0
M (kNm)